# Yuze Li

+1 (530) 364-8650 | wil018@ucsd.edu | San Diego, CA

## EDUCATION

***University of California, San Diego*** Expected Graduation: 06/2027
*Bachelor of Science in Mathematics - Computer Science* **UC-GPA: 3.885/4.0**
***University of California, Davis (Major: Data Science)*** 08/2023 - 09/2025

- ✧ Relevant Coursework
  - ✓ Data Structures, Math/Algorithms & Systems Analysis, System Programming and Software Tools, Theory of Computation, Algorithm Design & Analysis, Computer Organization, ML: Learning Algorithms, Computer Vision
  - ✓ Calculus, Linear Algebra, Vector Analysis, Mathematical Reasoning, Numerical Analysis, Differential Equations, Discrete Math & Graph Theory, Probability Theory, Math Statistics, Analysis of Variance, Regression Analysis, Statistical Data Science
- ✧ Award: UC Davis Deans' Honor List (4.0 Term-GPA), UC San Diego Provost Honors

## TECHNICAL PROFICIENCY

- ✓ Programming Language: Java, Python, C, C++, C#, SQL, R, JavaScript, HTML/CSS
- ✓ Framework: Spring Boot, Spring MVC, MySQL, Redis, Docker, Nginx
- ✓ Developing Tool: IntelliJ IDEA, PyCharm, RStudio, Git, Linux (CentOS)

## PROFESSIONAL EXPERIENCE

***Co-founder & VP of Engineering, Unifinity AI*** 07/2025 - Present
*AI-powered marketing platform – deliver strategy-content-publish-monitor operations for SMBs.*

- Architect a **microservices-based** backend using Java + Spring Boot, deploy on CentOS Linux with Docker, and integrate traffic routing with Cloudflare CDN and Nginx reverse proxy.
- Design and implement the core **online transactional pipeline** to organize real-time frontend user inputs into 23 tables, and feed to LLMs for generating AI-driven marketing strategies.
- Build a system for AI image generation using a **per-user token bucket limiter and round-robin scheduling**, ensuring equitable resource distribution across concurrent users under QPS quota.
- Stream strategies and social media posts with **Server-Sent Events pipelines**, integrating the Meta Graph API to schedule and publish content across Instagram, Facebook, and other platforms.
- Construct a secure file manager with role-based access control for stable uploads and metadata management, utilizing Google OAuth 2.0 authentication with reCAPTCHA protection.
- Define new feature by assessing feasibility, translating requests into actionable development plans, and coordinating engineering efforts across the team, driving the product from concept to launch.

***Software Development Intern, Compass Lessons (Remote)*** 07/2025 - 09/2025

- Integrated an invisible interactive object on the web frontend for bot-detection and request filtering to prevent web scraping and flood attacks, reducing malicious traffic by over 95%.
- Developed a **proxy-based function call layer** to standardize web traffic transaction pipeline with a unified logging interface, enhancing cross-team collaboration without ad-hoc coordination.
- Simplified debugging efforts and shortened average incident diagnosis time by ~25%, accelerating the overall development feedback cycle and improving system maintainability.

## DEVELOPMENT EXPERIENCE

***What4Dinner*** 02/2026 - Present

- Build a **prompt engineering pipeline** to convert unstructured social media cooking screenshots into structured recipes with automatic ingredient extraction, and design a Gemini based recommendation engine, offering suggestions based on user preferences and cooking history.
- Implement grouping features, enabling users to create Family Shared Workspaces, where members vote on meals and generate optimized grocery lists, fostering **dynamic information synchronization** and decision making within households.
- Develop backend services using Spring Boot, Spring Security, Google OAuth2, and JWT for secure authentication along with a responsive React frontend, utilize MySQL for persistent storage and Redis for caching, and deploy all services with Docker on CentOS.

***AdjacentTalk (Android, Windows)*** 08/2023 - 09/2024

*Full-featured cross-platform social app with real-time messaging, file sharing, contact management.*

- Handcrafted an **application-layer protocol** from scratch to manage client connections over TCP/IP raw sockets (socket, bind, listen, accept) with request/response handling, enabling interoperability and achieving low-latency, efficient data transmission without reliance on third-party frameworks.
- Designed a **distributed processing architecture** coordinating three independent servers to handle up to 1,000 concurrent users with sub-50 ms average message latency during internal beta testing, and implemented **load distribution** and failover mechanisms, enhancing service robustness.
- Utilized in-memory data structures to cache frequently accessed data, supporting up to 100,000 active chats through incorporating advanced executions, such as dynamically validating real-time inputs from users and triggering appropriate status code responses.
- Ensured consistent deployment of backend on CentOS 8 with MySQL and Docker, and tested server functionality using curl and a custom Bash test script that simulated 1,200+ concurrent-like requests across three endpoints, verifying correctness and error handling under load.

***StudyHelper (Available on Google Play Store)*** 07/2022 - 10/2022

- Managed the full development lifecycle of a utility app using Java and **Android SDK**, implementing core features including Memo, Phone Lock, and an advanced Scientific Calculator.
- Computed statistical functions (Normal and T distribution CDF) with accurate numerical handling.
- Ensured local data persistence with SQLite for reliable notes storage and retrieval, and handled device administration permissions and lock screen control through DevicePolicyManager.

## PROJECT EXPERIENCE

***Memory Allocator (vmalloc/vmfree)*** 10/2025

- Implemented a custom memory allocator with C, managing a simulated heap mapped and enforcing 16 byte alignment with **best fit allocation policy** to minimize fragmentation.
- Used pointer arithmetic and bitwise operations to manipulate block headers and compute block sizes, handling edge cases (e.g., zero size requests, allocation failures, NULL pointers).
- Validated correctness through passing a comprehensive test suite with zero memory errors, covering basic allocation, splitting, coalescing, heap images, and boundary conditions.

***Pish: Custom Unix Shell*** 09/2025

- Implemented a Unix like shell in C supporting both interactive and batch execution modes, building the core command loop with prompt display, line reading, and parsing to tokenize input.
- Managed process lifecycle with three built in commands, maintained persistent command history using file I/O, and performed robust error checking.